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Discrimination of EEG Related to motor imagery by combined wavelet energy feature with phase synchronization feature

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***Abstract*—For the problem of the EEG signals classification in Brain-computer interfaces (BCI), we proposed a method of feature extraction is that the combination of the wavelet energy feature and phase synchronization feature. In this paper EEG signals are transformed by means of discrete wavelet transform firstly, and then extracted 5 wavelet energy feature in different 5 frequency bands. We also obtained the instantaneous phase values based on Hilbert Transform, and extracted the phase synchronization feature through the method of Phase Locking Value (PLV). According to the distribution of the phase locking value in the time-domain, we obtained the best feature extracting time rage. Thus the final signal features are used as inputs for a Support Vector Machine (SVM) classifier. The result indicated the best feature extracting time rage is 4s~7s, and the accuracy of classification is 89.4%, providing an improvement comparing with only using wavelet energy feature.**

Keywords—Motor imagery; EEG; Wavelet transform; Phase synchronization

INTRODUCTION

Brain-computer interface (BCI) is an important international research and application, and the BCI based on motor imagery Electroencephalograph (EEG) is an important one [1]. Presently, one of the problems for the BCI technology is to find a good method of feature extraction and pattern classification algorithms [2]. The feature extraction method has a direct effect on the pattern classification [3].

Currently, extracting energy as the feature in the BCI based on motor imagery EEG is the primary feature extraction algorithm, the theory based on the brain electrophysiological phenomenon of ERD/ERS[4]. Researchers always use the wavelet coefficients as the energy of EEG. Tolic, M et al used the discrete wavelet transform coefficients as the feature to have a classification, mean classification accuracy for the recognition of all five tasks was 90.75% and mean classification accuracy for the recognition of two tasks was 99.87% [5]. Yeh, WL et al used wavelet transform and took the subjects' own EEG signals as the mother wavelet, choose Bayes linear discriminant analysis (LDA) as the classifier. Finally, the performance of classification had a improvement comparing with variance and fast Fourier transform (FFT) methods in feature extraction [6].

Energy as the primary feature in the BCI systems, has a good classification accuracy [7]. But the phase feature as the signal characteristics also, always be ignored by people. It contains some additional information outside of energy.

At present, the research that take the phase feature apply into BCI is a few.

This paper based on motor imagery brain electrical physiological principles, proposed a method of feature extraction is that the combination of the wavelet energy feature and phase synchronization feature. The EEG signals which include the data of C3 and C4 are transformed by means of discrete wavelet transform firstly, and then extracted 5 wavelet energy feature in different 5 frequency bands. We also obtained the instantaneous phase values of C3, C4 and Cz based on Hilbert Transform, and extracted the phase synchronization feature of C3-Cz and C4-Cz through the method of Phase Locking Value (PLV). According to the distribution of the phase locking value in the time-domain, we obtained the best feature extracting time rage. Thus the final signal features are used as inputs for a Support Vector Machine (SVM) classifier, and the accuracy of classification is 89.4%.

I. EXPERIMENTAL DATA

This paper use the BCI2003 competition data provided from the University of Graz in Austria as the experimental data. The experiment consists of online BCI system with feedback. Subjects imagine the right and left hand movement to control the movement of the cursor in the interface, timing scheme as shown in figure 1.

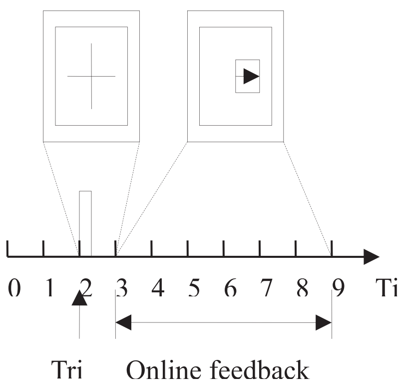


Figure 1 Timing scheme

Experiment divided into 7 groups, each group 40 times test, a total of 280 trials. Each test lasts 9s. Subjects be in quiet state in 0s~2s, at the central of the interface appear a symbol of "＋", and this symbol last 1s in 2s~3s, and then, appearing a symbol of right arrow or left arrow in the interface. At the same time, the system require subjects imagine the left or right hand movement based on the symbol of right arrow or left arrow in the interface, this process last 6s. The online system distinguish from two kinds of tasks, and apply adaptive AR model to extract the AR coefficient. The system acquire the classification results through Linear Discriminant analysis, and the result be provided to subjects as the feedback signal.

AgCl electrode was adopted in the experiment, and the data were acquired based on differential electrode from the international standard 10-20 lead system C3, C4 and Cz. The data contains of training data and test data, and each data had the left and right hand movement imagination 70 times each. The sampling frequency was 128Hz.

II. FEATURE EXTRACTION AND PATTERN CLASSIFICATION

A. *Feature Extraction*

1) The Wavelet Energy Feature

The principle of wavelet transform is applied wavelet function to represent or close the original signal, and the basic wavelet become the wavelets by translation and scaling [8]. We can get the projection of original signal through the framework of this function space. The signal in time domain can be expressed by time scale through multi-scale decomposition of wavelet transform. Mallat et al put forward the method to implement the algorithm of wavelet transform and its inverse transform by double-channel filters, we called this method "Wavelet fast algorithm" or "Mallat algorithm". $c_j(k)$, $d_j(k)$ are discrete approximation coefficients of resolution analysis, and $h0(k)$, $h1(k)$ is are two filters of dimension differential equation. The recurrence relation of $c_j(k)$, $d_j(k)$ as follows:

$$\begin{cases} c_{j+1}(k)=\sum_{n=-\infty}^{+\infty} c_j(n)h_0(n-2k)=c_j(k)*\bar{h}_0(2k) \\ d_{j+1}(k)=\sum_{n=-\infty}^{+\infty} c_j(n)h_1(n-2k)=c_j(k)*\bar{h}_1(2k) \end{cases} \tag{1}$$

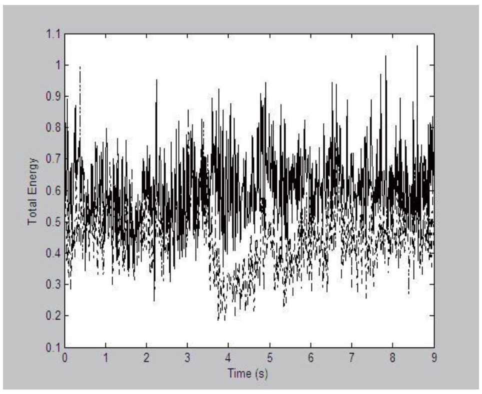
in the equation has: $\bar{h}(k)=h(-k)$.

Suppose x(i,j) is the jth times EEG data in ith times test, the calculation formula of energy of the jth times EEG data in totals of experiment as follows:

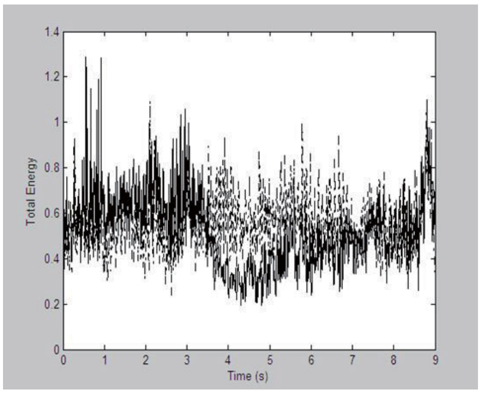
$$Pj=\sum_{i=1}^{N} x_{(i,j)}^{2} \tag{2}$$

N=70, Pj is the total sum of energy square of the jth data point[9]. The unit of the total energy is μV.

We can get the total energy data of imagine left or right hand movement each sampling point of C3 and C4 in 0s~9s based on the equation (2). We can see from the comparing of C3 andC4 in figure 2, when brain imagine the left and right hand movement, the wavelet coefficients have significant differences in 3s~8s. This provides the best period of time for feature extraction.



（a）the comparing between C3 and C4 of wavelet energy in left hand imagination



（b）the comparing between C3 and C4 of wavelet energy in right hand imagination

Figure 2 the comparing between C3 and C4 of wavelet energy in left and right hand imagination

The EEG signals in 3s~8s are decomposed based on the principle of wavelet decomposition. We used the Danbechies4 as the wavelet function. The EEG signals in 3s~8s are decomposed on 7 layers by Mallat algorithm. After wavelet decomposition, the corresponding frequency can be seen in table 1.

Table 1 the corresponding frequency band of wavelet decomposition Hz

signal	D1	D2	D3	D4
frequency	32~64	16~32	8~16	4~8
signal	D5	D6	D7	A7
frequency	2~4	1~2	0.5~1	0~0.5

As we all know, the frequency of power interference and the head electrical noise concentrated on more than 30Hz frequency band[10]. The frequency range of the δ rhythm is 0.97~3.91Hz, the frequency range of the θ rhythm is 3.91~7.8Hz, the frequency range of the α rhythm is 7.8~15.6Hz, the frequency range of the β rhythm is 15.6~31.25Hz. The corresponding frequencies D2, D3, D4, D5, D6 of wavelet decomposition contains the 4 kinds of rhythm. We use the 4 kinds of rhythm drain electrical signal energy representing the brain activity as part of feature vector. The energy E(x) calculation formula for EEG signals x(n) is:

$$E(x) = \sum_{n=0}^{N-1} \left| x(n) \right|^2 \qquad (3)$$

N is the length of x(n).

According to the equation (3), we calculate the 5 signals wavelet energy of D2, D3, D4, D5 and D6, and acquire E(x1), E(x2), E(x3), E(x4) and E(x5), as part of feature vector.

2) Phase Synchronization Feature

Motor control functional brain imaging suggest that the excitement of motor cortex can occur in multiple regions at the same time in the process of movement. The collaboration and integration between different regions in motor cortex can be embodied in the degree of synchronization between brain electrical signals in different areas [11].

At present, we know that the correlation of phase and amplitude correlation in signals were independent of each other. When the signal amplitude is unrelated, it's phase could keep synchronization [12]. The phase coupling between two signals can be available by Phase Locking Value (PLV). Phase Locking Value also called phase synchronization factor. It is defined as follows:

$$PLV = \frac{1}{N} \left| \sum_{t=1}^{N} \exp(j(\varphi_1(t) - \varphi_2(t))) \right| \qquad (4)$$

$\varphi_1(t)$ and $\varphi_2(t)$ represent of the instantaneous phase value of channel 1 and 2 at time t. The values of PLV is in range of 0~1. When the two signal phase in accordance with completes synchronization, the PLV value is 1, and when two signal phase completely out of sync, the PLV value is 0. Determination of PLV is stable of two channel signal rather than the size of the phase difference.

Instantaneous phase value can be calculated by Hilbert transform or wavelet transform. In this paper, we use the Hilbert transform to calculate the instantaneous phase. The Hilbert transform is described below:

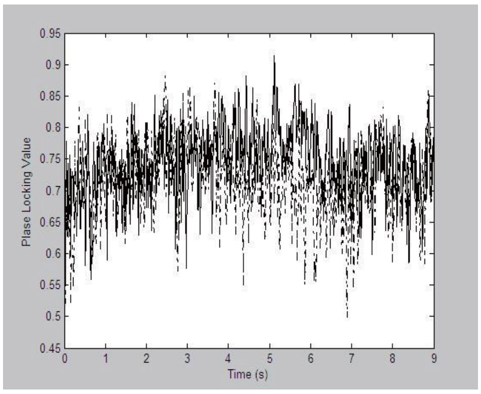
$$y(t) = \frac{1}{\pi} P \int_{-\infty}^{+\infty} \frac{x(\tau)}{t - \tau} d\tau \qquad (5)$$

The equation show a non-stationary signal x(t) become y(t) by Hilbert transform, the P is the Cauchy principal value [13]. Instantaneous phase can be calculated as follows:

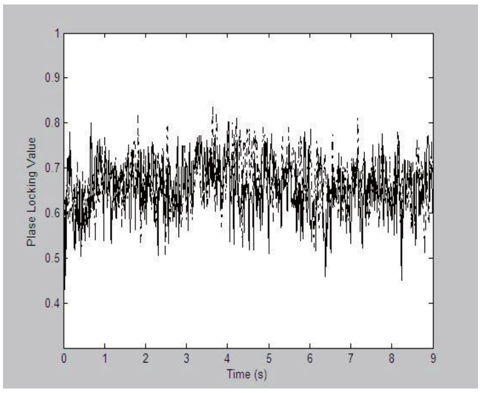
$$\varphi(t) = \arctan \frac{y(t)}{x(t)} \qquad (6)$$

According to the synchronization feature of EEG signal in the process of motor imagery, this paper selects PLV of two channels (C3-Cz and C4-Cz) as the phase synchronization feature.

We can see from the experimental sequence, motor imagination started at 3th seconds, and the corresponding phase synchronization in different areas of brain are differences in the time domain. In this paper, we find the best time of extracting phase synchronization feature through the training data. In figure 3 (a) and (b), solid lines represent the right hand imagine PLV distribution in the time domain, and dotted line represents the left hand imagine PLV distribution in the time domain. Each curve is constituted by PLV mean at each sample point in 70 single test of one imagine task. The figure 3 shows, phase synchronization of two imagine task is evident in the 4s~7s.



(a)the PLV temporal distribution of C3-Cz in two kinds of imagination (solid lines represent the right hand imagine PLV distribution and dotted line represents the left hand imagine PLV distribution)



(b)the PLV temporal distribution of C4-Cz in two kinds of imagination (solid lines represent the right hand imagine PLV distribution and dotted line represents the left hand imagine PLV distribution)

Figure 3 The PLV temporal distribution of C3-Cz and C4-Cz in two kinds of imagination (solid lines represent the right hand imagine PLV distribution and dotted line represents the left hand imagine PLV distribution)

Above all, each test extract 5 wavelet energy value from C3 and C4, and extract 2 phase synchronization feature from C3-Cz and C4-Cz. The feature vector is 12d.

B. *Pattern classification*

In this paper, we apply the radial basis kernel function as the kernel function of Support vector machine (SVM), and use the wavelet energy feature and phase synchronization feature as the feature vector to have a classification for the test data of EEG.

III. The Experimental Results

The EEG signals of 3 channels were collected in the experiment. We selected the C3 and C4 two channels of EEG signals on the feature of wavelet energy, and selected the PLV of C3-Cz and C4-Cz as the feature of phase synchronization. From the figure 3 we can see, phase synchronization of two imagine task is evident in the 4s~7s. In this paper, we extracted phase synchronization feature respectively in 1s~9s, 2s~7s, 3s~8s, 4s~7s and 4s~8s, and combined with wavelet energy feature. After recognized by classifier, the corresponding recognition accuracies are 81.2%, 84.3%, 86.7%, 89.4% and 88.1%, verifying the best time for phase synchronization feature extraction in 4s~7s. As shown in table 2.

Table 2 The Classification Accuracy of Phase Synchronization Feature Extraction Time

Time	Accuracy
1s~9s	81.2%
2s~7s	84.3%
3s~8s	86.7%
4s~7s	89.4%
4s~8s	88.1%

IV. DISCUSSION

Comparing with only using the wavelet energy as feature, the classification accuracy of combination feature has a obvious improvement, as shown in table 3.

Table 3 The Classification Accuracy of Different Feature Vector

Feature Vector	Accuracy
wavelet energy feature	85.70%
wavelet energy feature + phase synchronization feature	89.40%

The purpose of this paper is to research the improvement of phase synchronization feature for pattern classification. Comparing with the research of Tolic, M et al, the first in terms of experimental paradigm, we applied the competition data, and Tolic, M et al applied their own experimental data. Secondly, in addition to the wavelet energy feature, we combined the phase synchronization feature in this paper.

Comparing with the research of Yeh, WL et al, the difference is that used wavelet transform and took the subjects' own EEG signals as the mother wavelet, and the classification performance was greatly improved. And in this paper, we used the Mallat algorithm Danbechies4 wavelet as the mother wavelet.

Compared with the previous methods, the method of the combination of the wavelet energy feature and phase synchronization feature is relatively simple, and the effect is better. From the above analysis and result we can see, a higher classification accuracy can be acquired by using phase synchronization feature in a specific time. But the current weakness of this method is that the time of EEG signal treatment is a little longer. If applied to the online system may need to optimize the algorithm. The reason of longer processing time may be that we need extract two feature and combine it.

V. CONCLUSION

In this paper, we researched the method of the combination of the wavelet energy feature and phase synchronization feature to achieve the classification of EEG. We obtained the instantaneous phase values based on Hilbert Transform, and extracted the phase synchronization feature through the method of Phase Locking Value (PLV), and the test signal set is recognized by SVM. Experiments showed that, after the combination of the wavelet energy feature and phase synchronization feature, the classification accuracy is 89.4% in 4s~7s of phase synchronization feature, reached BCI level of competition. Data analysis showed that, phase synchronization feature contains additional information beyond the energy change. The performance of BCI system can be improved by the combination of the wavelet energy feature and phase synchronization feature.

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